AI Agents Document Operating System

Comprehensive User Guide & Practical Operational Manual

OFFICIAL VERSION 4.0

The Engineering Standard for Reliable AI Document Operations

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CHAPTER 1

Welcome: What is the AI Agents Document OS?

If you have ever asked an AI assistant to "read this PDF", "edit this spreadsheet", or "fix this PowerPoint presentation", you know the frustration. Standard LLM agents frequently hallucinate missing data, silently strip mathematical formulas in Excel, destroy corporate formatting, or generate corrupt files that fail to open.

AI Agents Document OS is your agent's reliable co-pilot. It provides a deterministic safety suite, forensic inspection engine, and quality auditor in one cohesive architecture. Your agent gains the specialized protocols needed to process real documents with zero data loss.

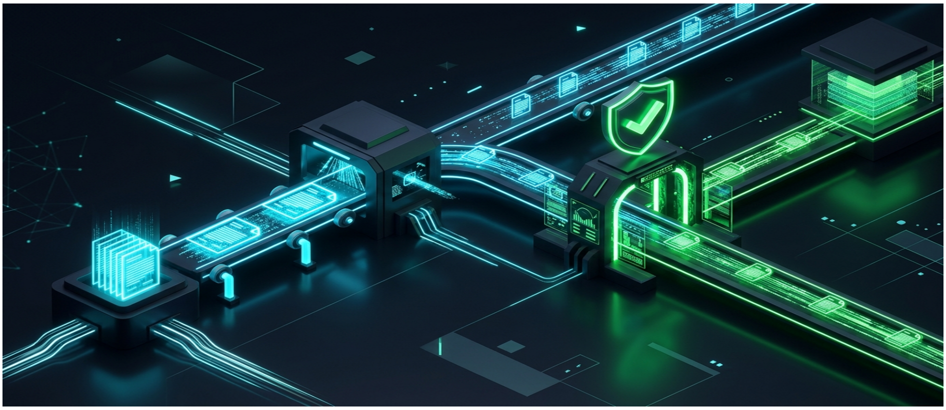
> AUTONOMOUS & NOVICE FRIENDLY
>
> You do not need to memorize terminal flags or write complex code. Simply speak to your agent in plain English. AI Agents Document OS triages file structures and applies safety protections automatically.

CHAPTER 2

Quickstart: Daily High-Fidelity Operations

When you collaborate with your agent, Document OS automatically detects target file types. Here are the 5 core everyday scenarios and how to prompt your agent:

Action	Sample Prompt	Document OS Protection
Read PDF	"Read the 2024 Audit and summarize revenue."	Identifies digital vs scanned layers; extracts tables cleanly.
Edit XLSX	"Update Q3 projections by 5%. Preserve formulas."	Loads workbook with formula graph guard; runs post-QA.
Convert Doc	"Convert proposal.docx into a presentation PDF."	Uses headless LibreOffice for 100% layout fidelity.
Slide Deck	"Build a 5-slide PowerPoint in 16:9 layout."	Applies master templates with word-wrap overflow guard.
Scanned OCR	"Extract vendor name and total from receipt.png."	Applies adaptive sharpening and deskewing before OCR.



CHAPTER 3

The 5-Stage Deterministic Safety Pipeline

Every document task passes through a 5-stage engineering pipeline to guarantee absolute fidelity:

Stage	Action	Tool / Helper	Fidelity Target
1. Triage	Structural inspection	inspect_doc.py	Detects digital text, counts formulas, checks aspect ratio.
2. Route	Protocol selection	document-router	Selects verified parser (openpyxl, pdfplumber, python-docx).
3. Execute	Deterministic runs	extract_doc.py convert_doc.py	Executes tested CLI commands without impromptu scripts.
4. Render	Visual verification	render_doc.py	Converts pages to high-res PNGs for vision inspection.
5. QA Audit	Integrity validation	qa_doc.py	Scans for formula errors (#REF!, #DIV/0!) and broken XML.

PIPELINE AUTOMATION

The 5-stage pipeline runs automatically. The Master Document Router triages structure and sequences the exact safe tools required for each file format.

CHAPTER 4

Workspace Architecture & Global Deployment

AI Agents Document OS is deployed in two complementary tiers:

- Local Workspace (.agents/plugins/document-os/): Directly integrated into your project repository. Shares deterministic rules with your entire engineering team.
- Global User Profile (~/.gemini/config/plugins/document-os/): Installed into your user profile so every agent session across all directories inherits Document OS capabilities.

Universal Agent Compatibility

Document OS strictly adheres to open agent specifications. Export seamlessly with one command:

```
.\export_cross_harness.ps1 -Target All
```

This establishes full compatibility with Google Antigravity, Claude Code, OpenCode CLI, and Cursor.

CHAPTER 5

CLI Command Reference & Automated Tooling

For CI/CD pipelines, automated scripts, or terminal workflows, use the deterministic CLI helpers:

Command	Description	Output / Guarantee
inspect_doc.py <file>	Deep structural triage.	JSON metadata (pages, sheets, formulas, fonts).
extract_doc.py <file> --tables	Extracts tabular data.	Clean Markdown tables ready for LLM processing.
convert_doc.py in.docx out.pdf	Headless conversion.	Pixel-perfect conversion via LibreOffice/Pandoc.
render_doc.py deck.pptx --outdir ./img	Renders slides/pages to PNG.	Visual artifacts for multi-modal agent inspection.
ocr_doc.py receipt.png --output txt	Tesseract adaptive OCR.	Pre-processed text extraction with contrast boost.
qa_doc.py model.xlsx	Pre/post integrity check.	Asserts 0 formula errors (#REF!, #DIV/0!).

CHAPTER 6

The Zero Unintentional Data Loss Guarantee

Document OS implements three non-negotiable architectural guarantees to safeguard your critical files:

1. Read-Only Default: The agent never mutates an original document in-place unless explicitly commanded. Output is written to distinct versions (e.g., budget_v2.xlsx).
2. Formula Integrity Guard: Dynamic calculation trees (`=SUM`, `=VLOOKUP`) are protected from being overwritten by static numerical values.
3. Mandatory Post-Execution QA: No operation is marked "Complete" until qa_doc.py validates file health, XML syntax, and rendering integrity.

ENGINEERED INTEGRITY

These rules are hardcoded into AGENTS.md, GEMINI.md, and document-safety.md, providing an unbreakable guardrail across all agent harnesses.

CHAPTER 7

Cross-Harness Freedom: Claude, OpenCode & Cursor

Seamlessly portable across all major AI coding platforms with identical safety guarantees:

Platform	Configuration Path	Integration Method
Google Antigravity	.agents/plugins/document-os/	Native plug-and-play.
Claude Code	~/.claude/skills/document-os/	Export via export_cross_harness.ps1.
OpenCode CLI	~/.config/opencode/skills/	Export via export_cross_harness.ps1.
Cursor IDE	.cursor/skills/document-os/	Export via export_cross_harness.ps1.

CHAPTER 8

Brand Identity, Web Design & Copywriting Skills

Version 4.0 introduces an integrated creative suite to ensure generated artifacts look and read like commercial-grade products:

- Brand Identity Skill: Royal Blue (#1A3A8F), Golden Yellow (#FFC107), and Navy (#0D1B4C) palette.
- Web Design Skill: Synthesizes designmd.ai tokens, neuform.ai depth, and dribbble.com card patterns.
- Copywriting Skill: Adopts the top-ranked skills.sh standard for high-converting headlines and CTAs.
- 1.6:1 Publication Standard: Enforces Amazon KDP book aspect ratio with hyperlinked navigation.

DOGFOODED STANDARD

This official guide is dogfooded proof: generated with the v4.0 PDF engine, embedded Google Fonts, clickable anchors, and the strict 1.6:1 height-to-width ratio.

About the Author

Ekpo Otu, Ph.D. is a Computer Professional, Full Stack Developer, Researcher, Author, and Lecturer dedicated to applying computational intelligence to solve real-world problems. His work focuses on autonomous agent architectures, reliable document engineering, and open-source tooling.

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